



Project Proposal Report

Streetify

Web-Based Transportation System

User Account Management (IT25102208 Nayanamina A.R.L.P.)	Ride Booking and Management (IT25102207 Dharmakeerthi W.A.C.B.)
Driver Trip Request Management (IT25102241 Senaka K.A.T.)	Payment and Receipt Management (IT25102225 Edirisinghe E.A.R.N.D.)
Review, Rating, and Feedback Management (IT25102193 Weerasingha W.A.M.B.)	Administration and System Control (IT25102240 Rammandalagedara R.V.S.)

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Introduction

Streetify is a web-based transport management platform developed as a second-year software engineering group project at SLIIT. Inspired by modern ride-hailing services like Uber, the system provides a seamless interface for passengers to book rides, track drivers in real-time, and complete secure payments — all within a single web application. The project is structured into six independent modules, each assigned to a dedicated team member, enabling parallel development while maintaining system cohesion through clearly defined API contracts and a shared database schema.

Problem Statement

- Inefficient & Unreliable Booking
- Communication Gaps between passengers and drivers
- Payment Transparency Issues
- Lack of Administrative Control

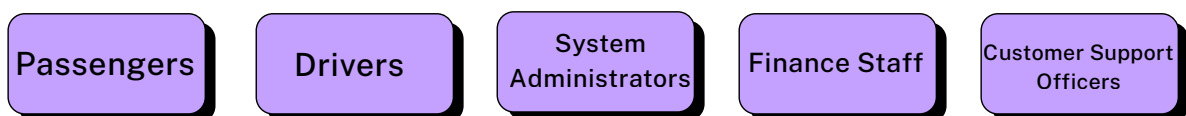
Solution

- User Onboarding & Verification
- Ride Request & Smart Matching
- Trip Management
- Multi-Mode Payment & Receipts
- Rating, Feedback & Dispute Handling
- Administrative & Financial Governance

Objectives & Goals

1. Simplify passenger ride booking and tracking.
2. Support drivers' transparent earnings.
3. Ensure secure multi-mode payment processing and digital receipts
4. Enable system administrators and finance teams to monitor operations and handle disputes. .

Target Users



Scope

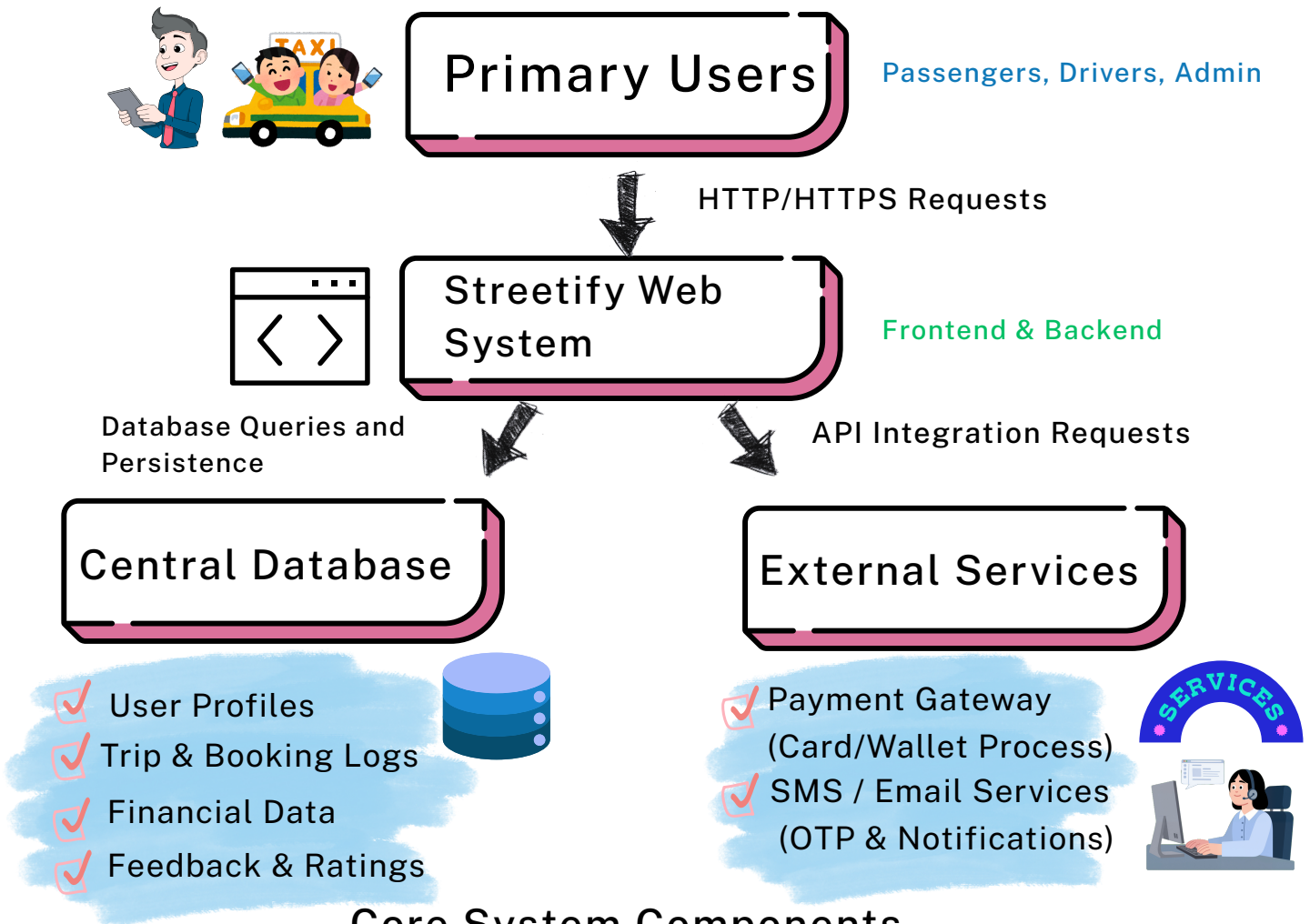
- Covers web application features for
- Account management,
 - Trip processing,
 - Fare estimation
 - Reviews/complaints,
 - Administrative controls.

Limitations

- Internet & Device Dependence
- Limited Operational Scope
- Manual Administrative Reliance

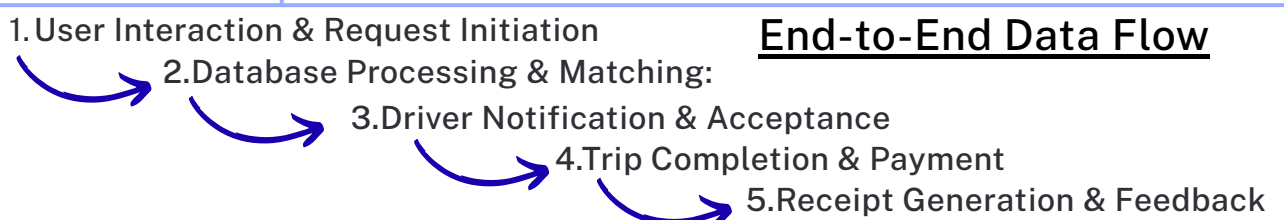


System Overview Diagram



Core System Components

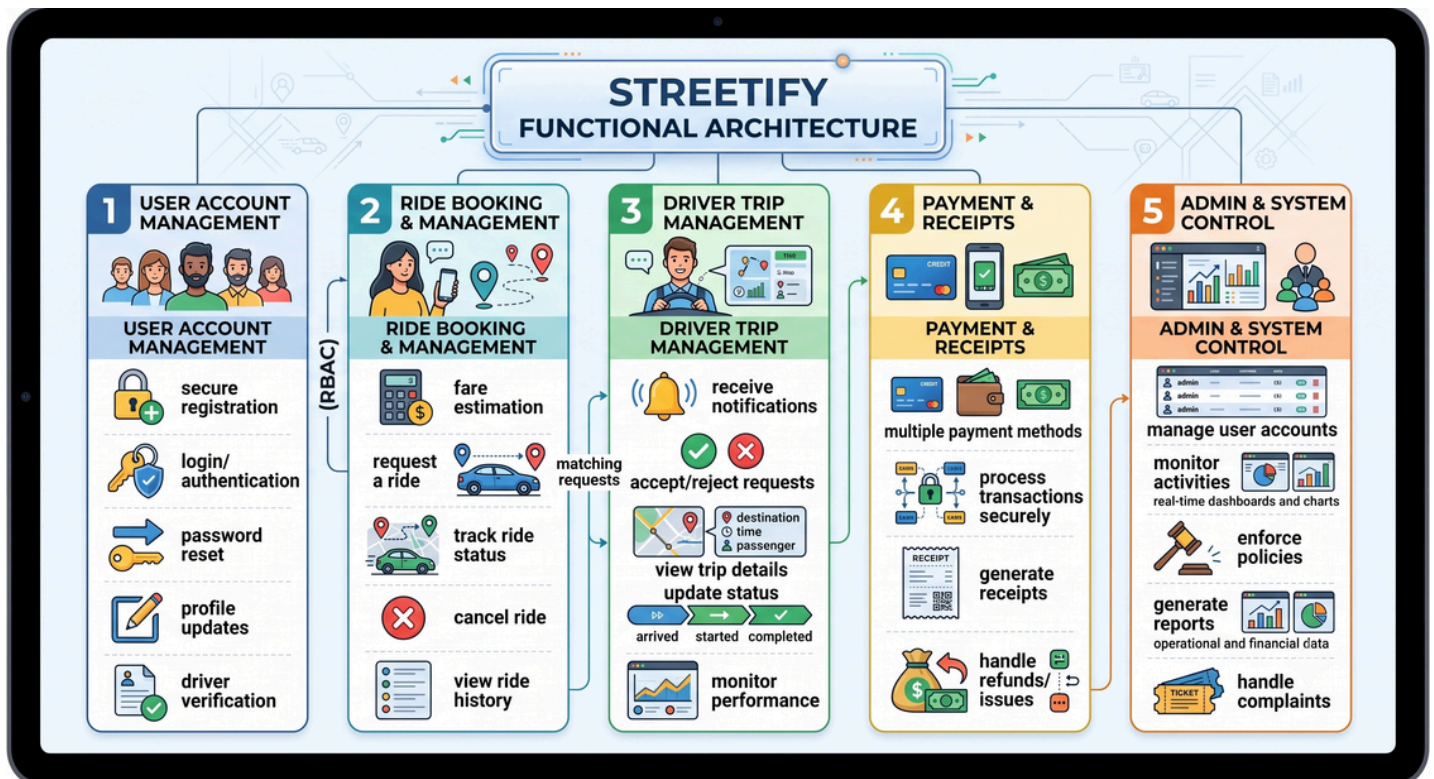
Primary Users (Clients):	Passengers booking rides, drivers receiving trip notifications, and administrators managing operations access the system through web browsers on mobile or desktop devices.
Streetify Web Application (Core Logic):	The central application handling core functional modules including User Account Management, Ride Booking, Trip Management, and Feedback Logging.
Central Database:	The primary data repository storing secure user accounts, active ride requests, transaction histories, system logs, and feedback entries.
External Services Integration:	<i>Payment Gateway Service:</i> Handles online card processing and digital wallet transactions securely. <i>Email & Notification Service:</i> Sends automated account verifications, password resets, booking updates, and digital receipts to users.





Functional Requirements

Functional requirements define the core actions and capabilities that the Streetify platform must support to ensure smooth operations for passengers, drivers, and administrative staff.



FR-01: User Account & Identity Management

Manages the lifecycle of user profiles across all roles (Passenger, Driver, System Administrator).

- **Registration & Auth:** Enables secure registration, login/logout, and password recovery via OTP/Email.
- **Driver Verification:** Allows administrators to upload, review, and verify driver licenses and vehicle registration documents.
- **Role-Based Access Control (RBAC):** Restricts access levels depending on whether the user is a passenger, driver, support officer, or system admin.

FR-03: Driver Trip Request Management

Provides drivers to receive, accept, and manage trip requests.

- **Notification Dispatch:** Notifies nearby drivers instantly of new incoming ride requests.
- **Accept / Reject Workflow:** Allows drivers to accept or decline trip requests with status updates sent to the passenger.
- **Trip Status Updates:** Enables drivers to update trip progress (e.g., Arrived, Trip Started, Completed) and view daily earnings.

FR-02: Ride Booking & Fare Management

Facilitates passenger ride requests from location selection to ride completion.

- **Fare Estimation:** Calculates estimated fares based on distance and route before booking confirmation.
- **Ride Request & Tracking:** Allows passengers to set pickup/drop-off points, submit requests, and track assigned driver details.
- **Cancellation & History:** Enables users to cancel pending requests and view complete past ride logs.

FR-04: Payment Processing & Receipt Generation

Manages transaction handling, fare calculations, and financial record-keeping.

- **Multi-Channel Payments:** Supports online credit/debit card transactions, digital wallets, and cash payments.
- **Automated Digital Receipts:** Generates and dispatches electronic invoices/receipts automatically upon trip completion.
- **Refund & Failed Transaction Handling:** Logs failed transactions and allows finance staff to process user refund requests.



FR-05: Administration, Review & System Control

Empowers administrative and customer support teams to maintain system safety, handle complaints, and oversee financial performance.

- Ratings & Feedback: Collects driver/passenger star ratings (1 to 5 stars) and written feedback.
- Dispute & Complaint Resolution: Enables support officers to investigate cancellation disputes and resolve customer complaints.
- System Monitoring & Reporting: Allows admins to suspend policy-violating accounts and export operational and financial reports.

Non-Functional Requirements

Non-functional requirements define the quality attributes, performance benchmarks, and operational constraints of the Streetify system. These ensure that the system is not only functional but also secure, efficient, user-friendly, and scalable.

1

Performance

Fast server response (< 2 seconds for requests) and high concurrency handling.

2

Usability

Intuitive, responsive web interface suitable for desktop and mobile web browsers.

3

Security

Encrypted user credentials, role-based authorization, and secure payment processing.

4

Reliability

High availability (24/7 uptime), automated database backups, and quick recovery protocols.

5

Maintainability

Modular software design adhering to Object-Oriented Principles for smooth future updates.



Major Stakeholders / Users



Primary Users:



1

Passenger

Registers, requests rides, tracks trips, completes payments, and leaves ratings/reviews.



2

Driver

Maintains driver profile, accepts/rejects trip requests, completes rides, and views earnings.



ADMINISTRATOR

3

System Administrator

Oversees platform security, verifies driver documentation, manages policy violations, and monitors system health.

Secondary Users:



1

Finance Staff / Manager

Monitors transactions, processes refunds, reviews driver commissions, and exports financial summaries.



2

Customer Support Officer

Handles cancellation disputes, investigates passenger complaints, and manages resolution tickets.



Six Major Functions

User Account Management

Create: Register new passenger/driver profiles.	Read: View user profile details, driver verification status, and login state.
Update: Edit contact information, password reset, update avatar.	Delete: Deactivate or archive user accounts.
Used By	Passengers, Drivers, System Administrators
What It Does	Manages registration, authentication, driver verification, and role-based access control across the system.
Rules & Dependencies	Driver accounts require admin verification, unique email/phone numbers, and rely on external email/SMS services for OTP and password recovery.
Expected Outcome	Secure authentication and strict role-based authorization across all user types.

Ride Booking & Management

Create: Create new ride requests with pickup/drop-off points.	Read: View ride tracking, estimated fare, driver/vehicle details, and past ride history.
Update: Modify route preferences or cancel active ride requests.	Delete: Clear completed or canceled ride records from active user views.
Used By	Passengers
What It Does	Handles the end-to-end booking process, distance/fare calculations, and ride monitoring.
Rules & Dependencies	Passengers need verified accounts; ride booking depends on driver availability in the region, and cancellations after acceptance may incur fees per policy.
Expected Outcome	A seamless, reliable ride-booking experience with transparent fare estimations.



Driver Trip Request Management

Create: Log trip acceptance records and daily trip entries.	Read: View incoming trip request details (passenger location, distance) and trip logs.
Update: Update trip status (Accepted, Arrived, Trip Started, Completed, Canceled).	Delete: Remove declined trip notifications from the driver dashboard.
Used By	Drivers
What It Does	Notifies drivers of nearby booking requests and enables real-time trip acceptance, execution, and tracking.
Rules & Dependencies	Drivers can accept requests only when Active/Available, triggered by Ride Booking, with repeated cancellations leading to admin alerts.
Expected Outcome	Efficient trip allocation and accurate status tracking for active drivers.

Payment & Receipt Management

Create: Process new payments (Card, Digital Wallet, Cash) and generate electronic receipts.	Read: Access transaction history, invoice details, driver earnings, and commission logs.
Update: Adjust payment status, apply promotional discounts, or issue refunds.	Delete: Flag and remove invalid/failed payment records.
Used By	Passengers, Drivers, Finance Manager / Staff
What It Does	Calculates total trip fares, processes payments across multiple channels, records driver commissions, and dispatches digital receipts.
Rules & Dependencies	Digital payments require gateway authorization, receipts auto-generate upon trip completion, and cash depends on driver's change availability.
Expected Outcome	Transparent, secure financial processing with automated receipt generation.



Review, Rating & Feedback Management

Create: Submit 1–5 star ratings, written reviews, and customer support complaints.	Read: View aggregate driver ratings, feedback logs, and customer complaint ticket statuses.
Update: Update complaint status (Pending, Under Investigation, Resolved).	Delete: Remove inappropriate reviews or resolved feedback entries.
Used By	Passengers, Drivers, Customer Support Officer
What It Does	Collects performance ratings, handles passenger/driver feedback, and provides ticket management for customer service support.
Rules & Dependencies	Ratings and reviews are allowed only after trip completion, linked to user accounts, with unresolved high-priority complaints triggering Customer Support alerts.
Expected Outcome	Continuous service quality control and efficient dispute resolution

Administration & System Control

Create: Generate system administrative reports (operational and financial dashboards).	Read: Monitor active user sessions, system audit logs, ride analytics, and financial statistics.
Update: Suspend or reactivate user/driver accounts due to policy violations.	Delete: Permanently delete inactive accounts or purge temporary system logs.
Used By	System Administrator, Finance Manager
What It Does	Provides administrative dashboards to oversee platform security, enforce policies, manage disputes, and export operational metrics.
Rules & Dependencies	Access is restricted to verified Admin/Finance users and relies on continuous data logging from Functions 1–5 to generate system metrics.
Expected Outcome	Comprehensive operational governance, high system security, and reliable data reporting.



Minor Functions

Common Supportive Functions

- User Login / Logout & Session Management.
- Password Reset & Account Recovery via Email/OTP.
- Profile Detail Updates (Contact Info, Avatar).



- In-App Notifications & Alerts for Trip/Payment Updates.
- General Help & Customer Support FAQs.



System Limitations / Constraints

Constraint Category	Specific Limitation	Impact
Connectivity	Mandatory Internet Access: Active network connection required on user devices.	Real-time features (booking, tracking, payments) are unavailable offline.
Regional Scope	Urban Focus: Initial deployment restricted to Sri Lankan urban/suburban zones.	Matching algorithms and service availability are limited to selected municipal boundaries.
Operational	Cash Availability: Cash payments depend on driver availability.	Drivers may decline cash trips if exact physical change is unavailable; digital payment is encouraged.
Governance	Manual Administrative Review: Policy enforcement and driver verification require human admin approval.	Dispute resolution and account approvals depend on admin response times.
Architecture	Web Application Platform: Built as a responsive web app rather than a native mobile application.	Native background GPS tracking is limited by browser permissions.



Simple Timeline

Timeline	Phase / Key Deliverable	Details & Planned Milestones
Weeks 3-4	Requirement Gathering	Client interviews, user roles, core function identification.
Weeks 5-6	System & Architecture Design	Class diagrams, sequence diagrams, UI wireframes.
Weeks 7-8	Database & Environment Setup	Schema design, database setup, initial CRUD endpoints.
Weeks 9-10	Core Development	Building 6 major functions and minor utility functions.
Weeks 11	System Testing	Unit testing, integration testing, bug fixes.
Weeks 12	Client Feedback & Refinement	User acceptance testing, UI polishing.
Weeks 13	Deployment Preparation	Server setup, documentation finalization, report review.
Weeks 14	Final Presentation & Submission	Final project delivery and viva presentation.



Conclusion

Streetify is a web-based ride-hailing system for SwiftRide Lanka (Pvt) Ltd, integrating six core modules—accounts, booking, driver requests, payments, reviews, and administration—into one platform, with defined roles for passengers, drivers, admins, finance, and support to ensure efficient, transparent urban transit.

Passenger Value	Driver Benefits	Business Control	Technical Feasibility
Transparent Fares	Fair Trip Logs	Real-Time Logs	Modular OOP Design
Instant Booking	Clear Earnings	Financial Audits	Scalable Web System

Why the Project is Worth Implementing...

- **Enhanced User Experience:** Passengers gain access to instant ride bookings, transparent pre-trip fare estimations, live ride tracking, and flexible multi-channel payment processing (card, wallet, cash) accompanied by digital receipts.
- **Empowering Transportation Service Providers:** Drivers benefit from real-time trip notifications, automated trip status updates, transparent daily earning reports, and reduced idle time between bookings.
- **Operational Oversight & Financial Governance:** System administrators and finance teams receive centralized web tools to enforce platform safety policies, monitor driver verifications, resolve cancellation disputes, and generate detailed operational and financial analytical reports.

Development Methodology

1. Object-Oriented & Modular Architecture
2. Clear Task Allocation & Individual Ownership
3. Disciplined Milestone Execution
4. Rigorous Quality Assurance & Refinement

Through this structured execution strategy, Streetify delivers a highly feasible, scalable, and commercially valuable solution for Sri Lanka's transportation ecosystem.

Thank You